Lecture 10: Linear Regression

Math for Data Science · 2026

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Agenda

- Linear Regression
- Probabilistic View
- Taylor Series

Why linar regression?

- Easy and informative
- Many fancy statistical learning approaches can be seen as generalizations or extensions of linear regression
- A useful jumping-off point

Linear Regression

Linear Regression Can Tell You A Lot

Imagine you're a statistical consultant hired to advise a company on their marketing strategy. They might wish to know:

1. Is there a relationship between advertising budget and sales?
2. How strong is the relationship between advertising budget and sales?
3. Which media are associated with sales? (e.g. TV, radio, newspaper)
4. How large is the association between each medium and sales?
5. How accurately can we predict future sales?
6. Is the relationship between advertising and sales linear?
7. Is there synergy among the advertising media?

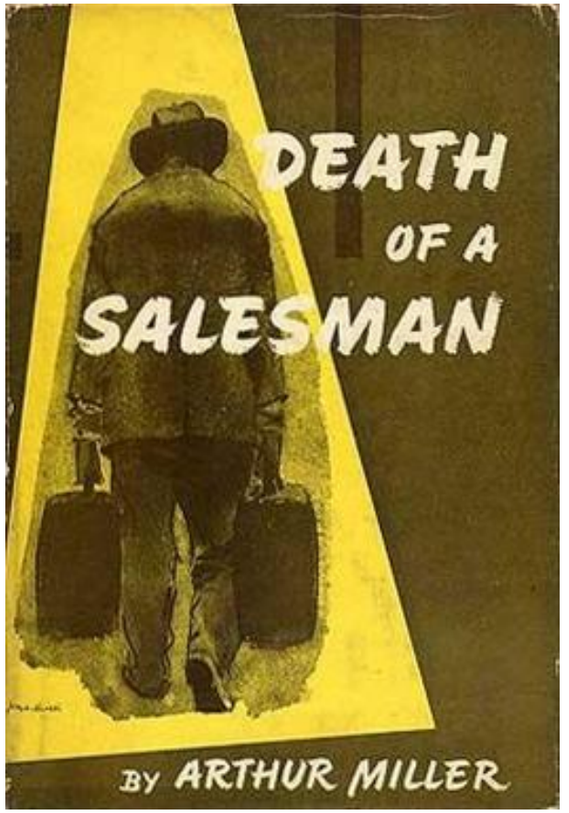
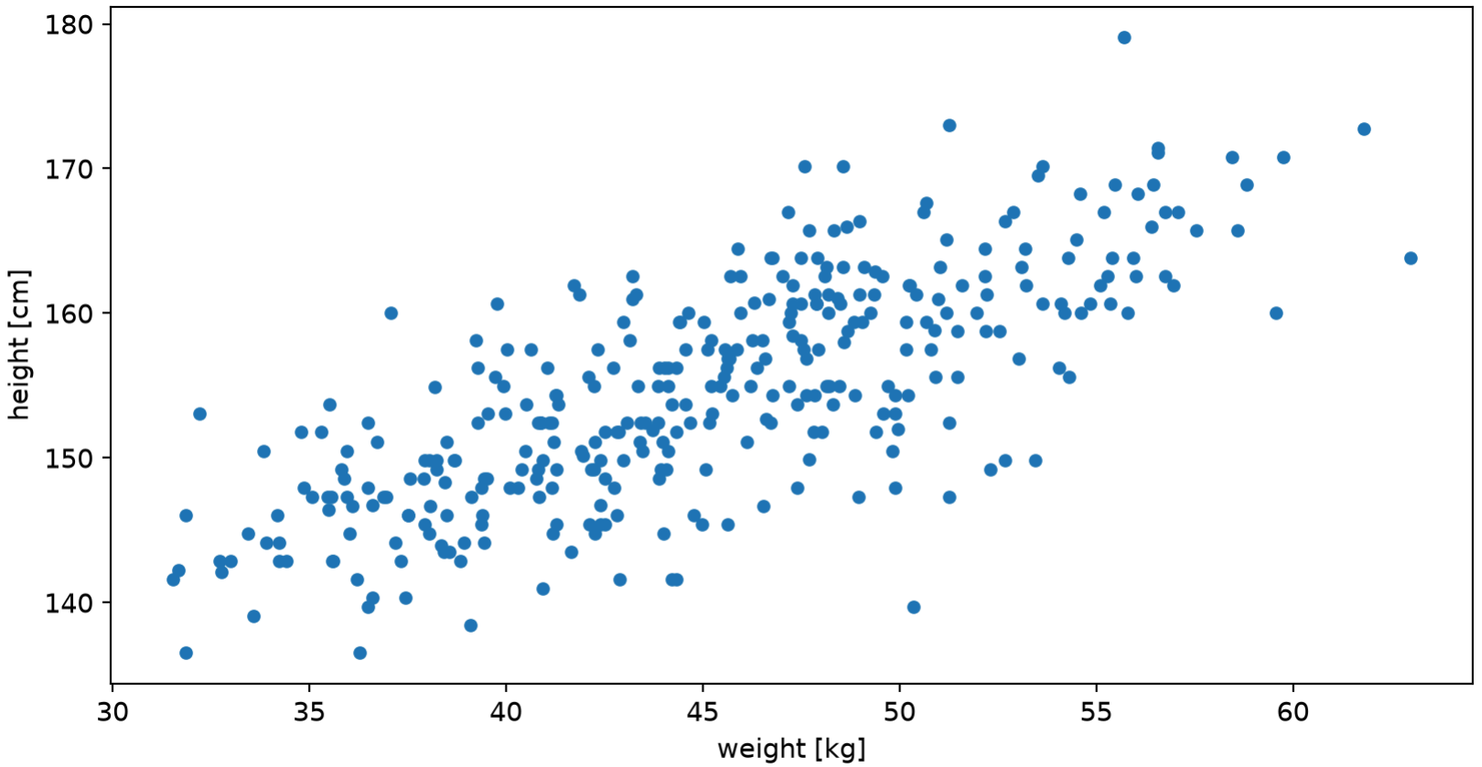


Image: book cover of Arthur Miller, *Death of a Salesman*, via Wikipedia (fair use).

Example: What is the relationship between weight and height?



Data: Howell census data of the Dobe !Kung San, from R. McElreath, *Statistical Rethinking* (`rethinking` package).

Linear Regression as optimization

The linear regression optimization problem is to minimize the residual sum of squares:

$$\mathrm{RSS} = \sum_{i=1}^{n} \varepsilon_i^2 = \sum_{i=1}^{n} (y_i - f(x_i))^2 = \sum_{i=1}^{n} (y_i - (\beta_0 + \beta_1 x_i))^2$$

Set up the minimization problem

$$\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} \varepsilon_i^2$$

and solve

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$
$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

the ^ denotes the estimate, the bar the mean

From One Covariate to Many: matrix form

$$\begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p} \\ 1 & x_{21} & x_{22} & \dots & x_{2p} \\ \dots & & & & \\ 1 & x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \dots \\ \beta_p \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \dots \\ \varepsilon_n \end{bmatrix}$$

more compactly

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

- $Y \in \mathcal{R}^{n \times 1}$ is called the response vector
- $X \in \mathcal{R}^{n \times p+1}$ is called the design matrix
- $\boldsymbol{\beta} \in \mathcal{R}^{p+1 \times 1}$ is our vector of coefficients
- $\boldsymbol{\varepsilon} \in \mathcal{R}^{n \times 1}$ is our vector of error terms

From One Covariate to Many: matrix form

Now, let's write the sum of squared error minimization problem in matrix form:

$$\sum_{i=1}^{n} \varepsilon_i^2 = \begin{bmatrix} \varepsilon_1 & \varepsilon_2 & \dots & \varepsilon_n \end{bmatrix} \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \dots \\ \varepsilon_n \end{bmatrix} = \boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon}$$

With a bit of manipulation:

$$\begin{aligned} \boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon} &= (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) \\ &= \mathbf{Y}^T \mathbf{Y} - \mathbf{Y}^T \mathbf{X}\boldsymbol{\beta} - \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{Y} + \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{X}\boldsymbol{\beta} \end{aligned}$$

The first derivative

Now, take the first derivative with respect to the entire $\boldsymbol{\beta}$ vector in one shot:

$$\frac{\partial}{\partial \boldsymbol{\beta}} \boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon} = \frac{\partial}{\partial \boldsymbol{\beta}} (\mathbf{Y}^T \mathbf{Y} - \mathbf{Y}^T \mathbf{X} \boldsymbol{\beta} - \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{Y} + \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{X} \boldsymbol{\beta}) =$$

$$\frac{\partial}{\partial \boldsymbol{\beta}} \mathbf{Y}^T \mathbf{Y} - \frac{\partial}{\partial \boldsymbol{\beta}} \mathbf{Y}^T \mathbf{X} \boldsymbol{\beta} - \frac{\partial}{\partial \boldsymbol{\beta}} \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{Y} + \frac{\partial}{\partial \boldsymbol{\beta}} \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{X} \boldsymbol{\beta}$$

The solution

As an exercise (in labs):

- Use the handy Matrix Cookbook (Section 2.4, Derivatives of Matrices, Vectors and Scalar Forms) to show that the equation on the previous slide reduces to:

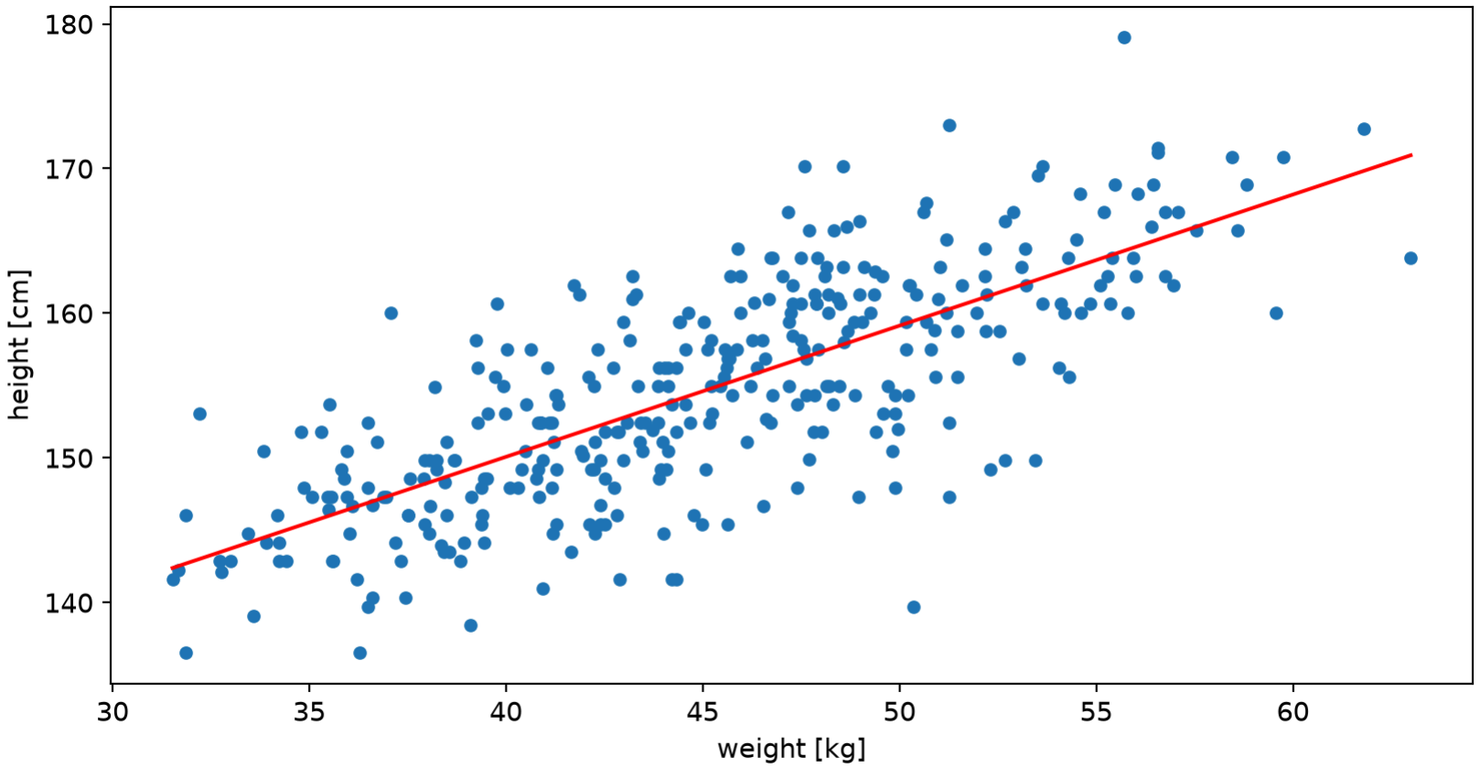
$$\frac{\partial}{\partial \boldsymbol{\beta}} \boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon} = -2\mathbf{X}^T\mathbf{Y} + 2\mathbf{X}^T\mathbf{X}\boldsymbol{\beta}$$

- Set the equation equal to 0 and solve for $\hat{\boldsymbol{\beta}}$. Show that:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}$$

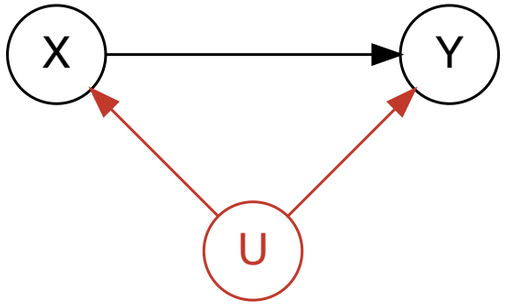
$Y \in \mathcal{R}^{n\times 1}, X \in \mathcal{R}^{n\times p+1}, \boldsymbol{\beta} \in \mathcal{R}^{p+1\times 1}, \boldsymbol{\varepsilon} \in \mathcal{R}^{n\times 1}$

Example: What is the relationship between weight and height?



Data: Howell census data of the Dobe !Kung San, from R. McElreath, *Statistical Rethinking* (`rethinking` package).

Causal Interpretation of Regression



- Causal question: what is the effect of getting a Hertie School degree (X) on your earnings (Y)?
- Confounding: your excellent prior academic record (U) made you competitive for admission to the Hertie School (X), and it also makes you competitive for high-paying jobs (Y).
- Naive regression: $Y = \beta_0 + \beta_1 X$
- Regression with causal interpretation: $Y = \beta_0 + \beta_1 X + \beta_2 U$

More powerful than you might think

We can create features that are arbitrary functions of the predictors, e.g.

- polynominals $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \dots$
- splines $y = \beta_0 S_1(x) + \beta_1 S_2(x) + \dots$
- trigonometric $y = \beta_0 + \beta_1 \sin(x) + \beta_2 \cos(x) + \beta_3 \sin(2x) + \beta_4 \cos(2x) + \dots$

This allows allows for interactions between predictors, such as

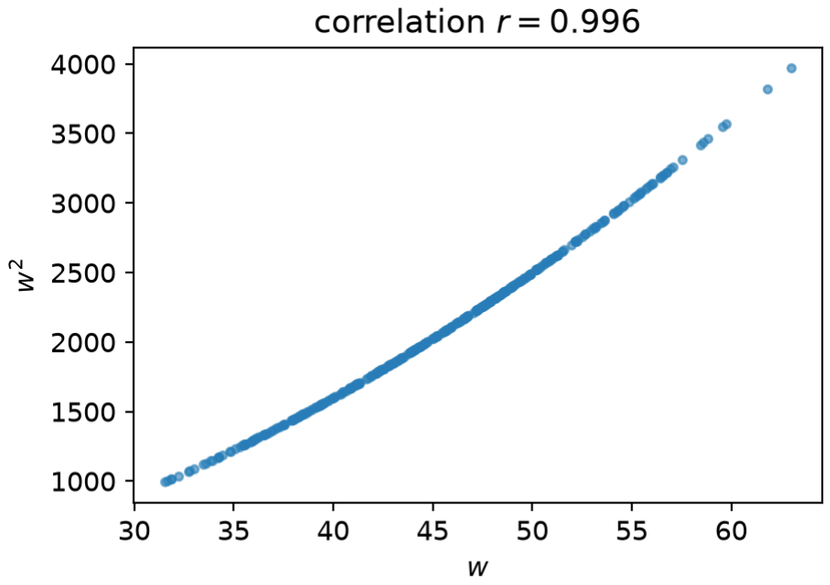
- $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$

It is important to note that the above is still linear in the β coefficients!

Adding mode feature does not always help

If two columns of $\mathbf{X}$ are linearly dependent, $\mathbf{X}^T\mathbf{X}$ is singular and $\hat{\boldsymbol{\beta}}$ does not exist.

If they are nearly dependent, the inverse exists but has an eigenvalue that is huge. Noise along the direction of that eigenvalues gets amplified and we get noisy results.



For example: Add a polynomial feature to the height-weight fit: the columns w and w^2.

Away from zero, w^2 is almost a straight line in w.

Nearly collinear columns make an eigenvalue d_i of $\mathbf{X}^T\mathbf{X}$ close to zero. The inverse $(\mathbf{X}^T\mathbf{X})^{-1} = \mathbf{V}\mathbf{D}^{-1}\mathbf{V}^T$ has eigenvalues $1/d_i$, so that one becomes huge.

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}$$

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\varepsilon}$$

Equally Good Fits, Different Coefficients

Fit height on w alone, and on w and w^2, for five random resamples of the data (drawn with replacement):

	w only: $\hat{\beta}_w$	R^2	w, w^2: $\hat{\beta}_w$	$\hat{\beta}_{w^2}$	R^2
1	0.91	0.615	0.68	+0.0026	0.615
2	0.95	0.627	0.50	+0.0049	0.628
3	0.95	0.595	1.04	-0.0009	0.595
4	0.90	0.594	1.11	-0.0022	0.594
5	0.88	0.529	0.49	+0.0043	0.530
std	**0.04**		**0.47**	**0.0051**	

Adding w^2 does not improve the fit, but $\hat{\beta}_w$ now swings by a factor of two and the curvature flips sign. The extra coefficient is fitted noise.

std over 1000 resamples

Ridge regression

The noise makes $\|\hat{\boldsymbol{\beta}}\|$ too large, so penalise it:

$$\hat{\boldsymbol{\beta}}_{\text{ridge}} = \arg\min_{\boldsymbol{\beta}} \; \boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon} \; + \; \lambda \, \boldsymbol{\beta}^T \boldsymbol{\beta}$$

The penalty adds one term to the gradient:

$$\frac{\partial}{\partial \boldsymbol{\beta}} \left(\boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon} + \lambda \, \boldsymbol{\beta}^T \boldsymbol{\beta} \right) = -2\mathbf{X}^T \mathbf{Y} + 2\mathbf{X}^T \mathbf{X} \boldsymbol{\beta} + 2\lambda \boldsymbol{\beta} = 0$$

$$\hat{\boldsymbol{\beta}}_{\text{ridge}} = \left(\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I} \right)^{-1} \mathbf{X}^T \mathbf{Y}$$

$\mathbb{E} \, \|\hat{\boldsymbol{\beta}}\|^2 = \|\boldsymbol{\beta}\|^2 + \sigma^2 \sum_i \frac{1}{d_i}$: the noise always lengthens $\hat{\boldsymbol{\beta}}$, and a $d_i \approx 0$ lengthens it a lot.

Standardise the predictors first: a coefficient's size depends on its variable's unit, so a variable measured in small numbers needs a large coefficient, which $\lambda \boldsymbol{\beta}^T \boldsymbol{\beta}$ then penalises hardest.

Adding $\lambda \mathbf{I}$ shifts every eigenvalue of $\mathbf{X}^T \mathbf{X}$ from d_i to $d_i + \lambda$, so the inverse's eigenvalues are capped at $1/\lambda$. Even for exactly collinear columns ($d_i = 0$), the matrix can be inverted.

Shrinkage Methods

- Ridge: $\min_{\beta} \sum_{i=1}^{n} \varepsilon_i^2 + \lambda \sum_p \beta_p^2$

  Shrinks all coefficients but keeps them. Use when many features contribute a little or come in correlated groups.
- Lasso: $\min_{\beta} \sum_{i=1}^{n} \varepsilon_i^2 + \lambda \sum_p |\beta_p|$

  Sets some coefficients to exactly zero. Use when you expect only a few features to matter.
- Elastic Net: weighted mixture of both

In short: Ridge adds stability, Lasso adds sparsity, Elastic Net combines the two.

$\mathrm{RSS} = \sum_{i=1}^{n} \varepsilon_i^2$

See ISL Ch. 6.2

Elastic Net - Explained

Probabilistic View

The generative model

Our model

$$y_i = f(x_i) + \varepsilon_i, \quad \varepsilon_i \overset{\text{i.i.d}}{\sim} \mathcal{N}(0, \sigma^2)$$

in the linear approximation

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i = \hat{y}_i + \varepsilon_i$$

Equivalently, $y_i \mid x_i \sim \mathcal{N}(\beta_0 + \beta_1 x_i, \sigma^2)$.

With many covariates, $y_i = \mathbf{x}_i^T \boldsymbol{\beta} + \varepsilon_i$, where $\mathbf{x}_i^T$ is the i-th row of the design matrix $\mathbf{X}$.

Assumptions:

1. Linearity: $\mathbb{E}[y_i \mid x_i] = \beta_0 + \beta_1 x_i$.
2. Gaussianity of the errors.
3. Homoscedasticity: the same σ^2 for every i.
4. Independence across observations.

Maximum likelihood

By independence, the likelihood factorizes:

$$L(\boldsymbol{\beta}, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \mathbf{x}_i^T \boldsymbol{\beta})^2}{2\sigma^2} \right).$$

Log-likelihood:

$$\ell(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \underbrace{\sum_{i=1}^{n} \underbrace{(y_i - \mathbf{x}_i^T \boldsymbol{\beta})^2}_{\varepsilon_i^2}}_{\text{RSS}}.$$

Maximize over $\boldsymbol{\beta}$.

$$\arg\max_{\boldsymbol{\beta}} \ell(\boldsymbol{\beta}, \sigma^2) = \arg\min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2.$$

Maximum likelihood

For a linear model with an assumed isotropic Gaussian likelihood, the maximum-likelihood estimate of $\boldsymbol{\beta}$ is the least square estimate.

$$\arg\max_{\boldsymbol{\beta}} \mathscr{\ell}(\boldsymbol{\beta}, \sigma^2) = \arg\min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2.$$

Note that the estimator of σ, $\hat{\sigma}^2_{\mathrm{MLE}} = \frac{\mathrm{RSS}}{n}$ is biased. For large n the bias vanishes.

Regression gives $s^2 = \mathrm{RSS}/(n-p-1)$. Fitting $p+1$ parameters uses up $p+1$ directions: the noise lying in those directions gets absorbed into $\hat{y}$ and disappears from the residuals. Only $n-p-1$ directions of noise remain to estimate σ^2 from.

Bayes

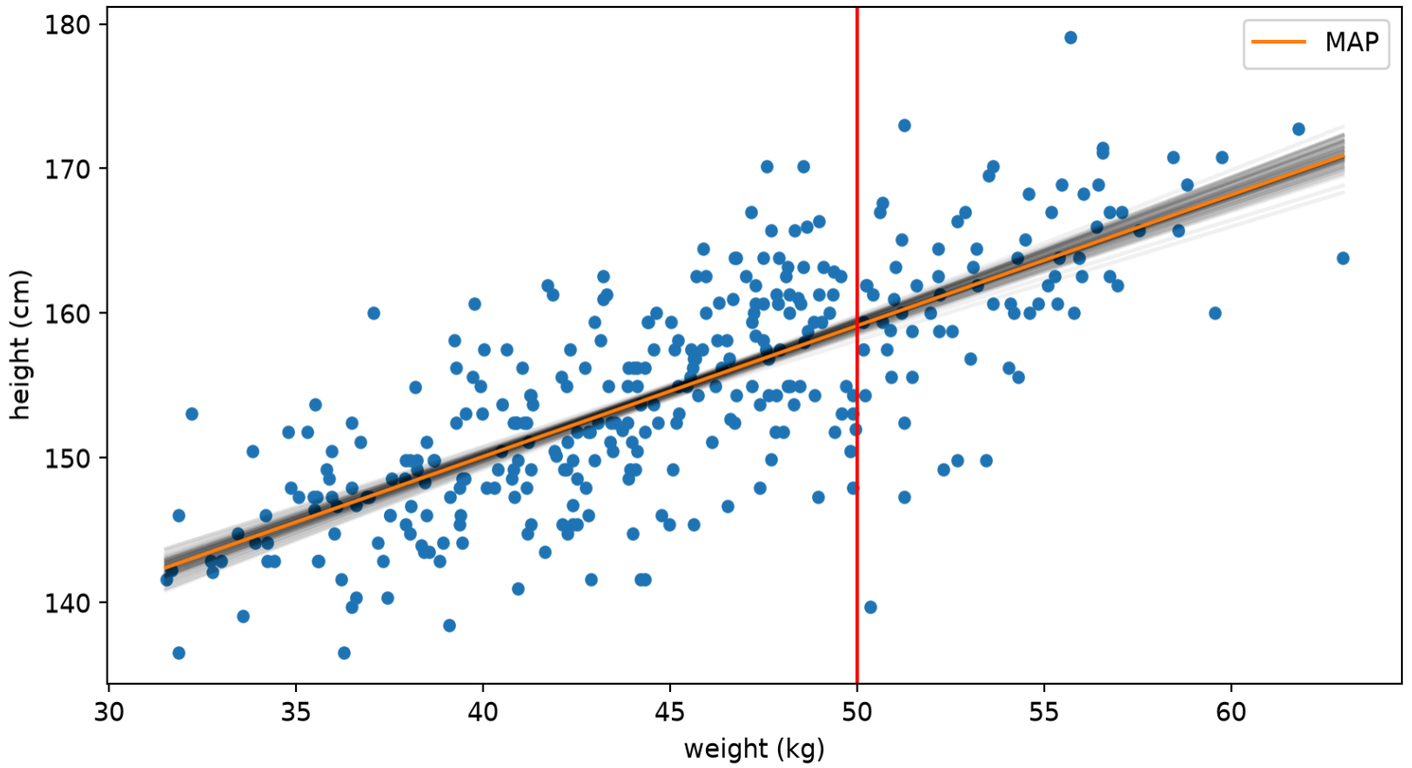
$$
\begin{aligned}
h_i &\sim \mathcal{N}(\mu_i, \sigma) && \text{[likelihood]} \\
\mu_i &= \beta_0 + \beta_1 (x_i - \bar{x}) && \text{[linear model]} \\
\beta_0 &\sim \mathcal{N}(178, 20) && \text{[prior]} \\
\beta_1 &\sim \mathcal{N}(0, 10) && \text{[prior]} \\
\sigma &\sim \mathcal{U}(0, 50) && \text{[prior]}
\end{aligned}
$$

The MAP estimate minimises $\frac{1}{2\sigma^2}\mathrm{RSS} - \log p(\boldsymbol{\beta})$, so the prior is the penalty:

- a Normal prior on $\boldsymbol{\beta}$ gives Ridge
- a Laplace prior gives Lasso
- a flat prior gives least squares.

$\mathcal{U}$ Uniform, $\mathcal{N}$ Normal, Laplace $p(\beta) \propto e^{-|\beta|/b}$

Bayes

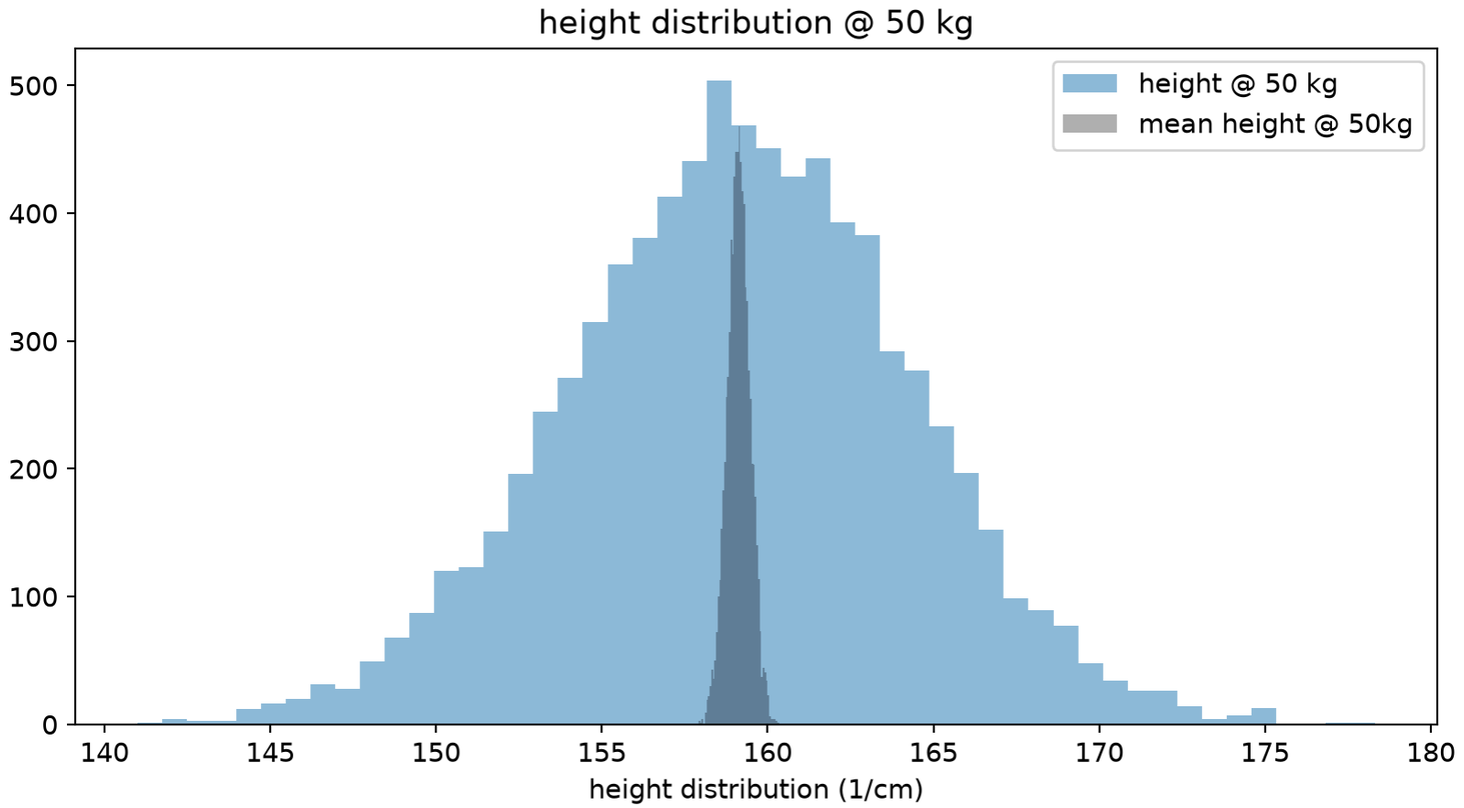


100 samples from the posterior, see also McElrath Fig. 4.7

Bayes: Prediction interval

Two kinds of uncertainty.

1. The uncertainty in the estimate of the mean.
2. The uncertainty of the sampling process (σ).



see McElrath Fig. 4.10

Taylor Series

Approximating a function with a polynomial

Suppose you have some ugly function $f(x)$. It's difficult to differentiate, integrate, manipulate, maximize, or even write down…

But luckily, let's assume it has a power series representation at some point $x = a$. In other words, suppose there is some polynomial function $p(x)$ such that:

$$p(x) = f(x) \text{ at } x = a$$

And recall that our polynomial function $p(x)$ looks like this:

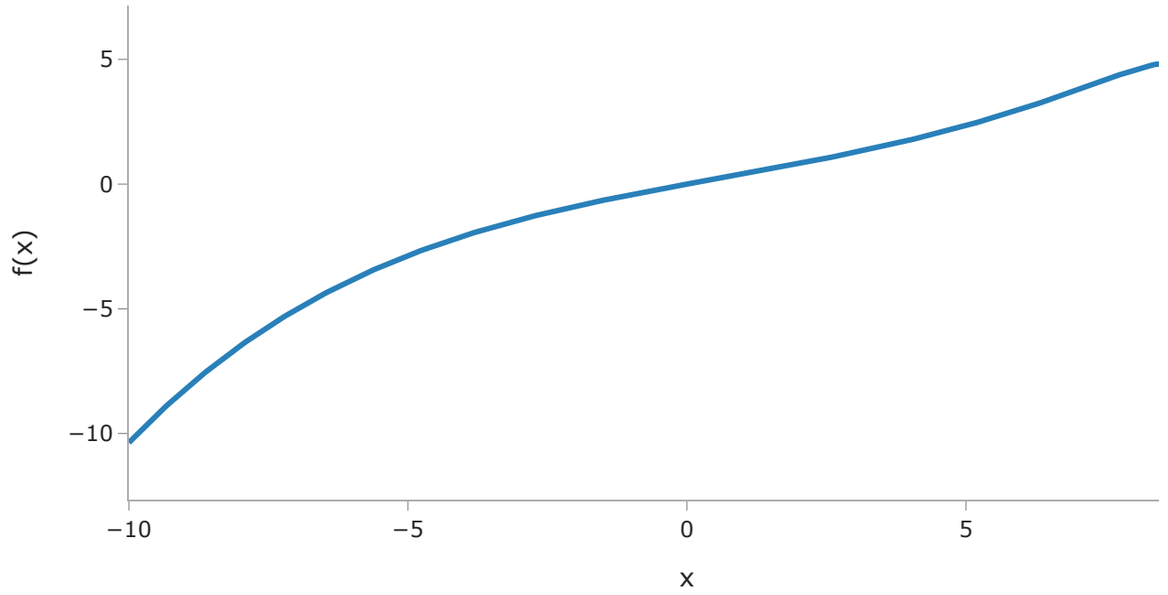
$$p(x) = \sum_{j=0}^{\infty} c_j (x - a)^j$$

Question: What should $p(x)$ be for a specific function $f(x)$?

Approximating a function with a polynomial at $x = 0$

Now, if we want $p(x)$ to represent $f(x)$ at $x = 0$, what conditions would we like to hold?

- $p(0) = f(0)$
- $p'(0) = f'(0)$
- $p''(0) = f''(0)$
- $p'''(0) = f'''(0)$
- $\dots$



We can solve for the values of c that make these conditions hold.

First conditions $p(0) = f(0)$ and $p'(0) = f'(0)$

Remember, what is $p(x)$? It's:

$$p(x) = c_0 + c_1 x + c_2 x^2 + \dots$$

So, $p(0) = c_0$. Then our first condition is: $\Rightarrow c_0 = f(0)$

Ok, so what is $p'(x)$? It's:

$$\begin{aligned} p'(x) &= \frac{d}{dx}\left(c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \dots\right) \\ &= c_1 + 2c_2 x + 3c_3 x^2 + 4c_4 x^3 \dots \end{aligned}$$

So, $p'(0) = c_1$. And we have our second condition: $\Rightarrow c_1 = f'(0)$

Keep Going!

Our third condition: $p''(0) = f''(0)$

What is $p''(x)$? It's:

$$\begin{aligned} p''(x) &= \frac{d}{dx}\left(c_1 + 2c_2 x + 3c_3 x^2 + 4c_4 x^3 \dots\right) \\ &= 2c_2 + 6c_3 x + 12c_4 x^2 \dots \end{aligned}$$

Then, $p''(0) = 2c_2$, giving us our third condition: $2c_2 = f''(0) \Rightarrow c_2 = \frac{1}{2} f''(0)$

What Have We Derived?

If you kept doing this, you would see a pattern:

$$p(x) = \underbrace{f(0)}_{c_0} + \underbrace{f'(0)}_{c_1}x + \underbrace{\frac{1}{2!}f''(0)}_{c_2}x^2 + \underbrace{\frac{1}{3!}f'''(0)}_{c_3}x^3 \dots$$

And, generalizing from $x = 0$ to $x = a$,

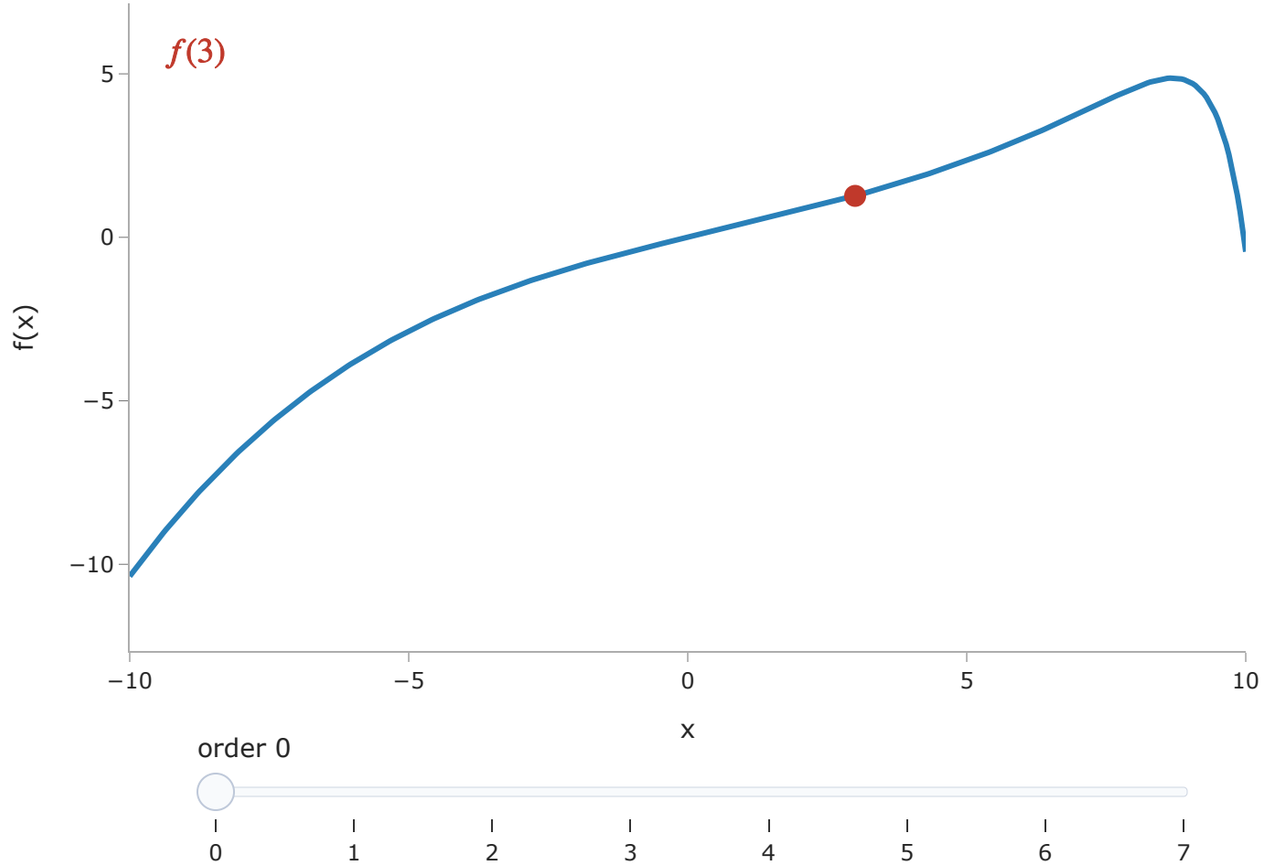
$$p(x) = \underbrace{f(a)}_{c_0} + \underbrace{f'(a)}_{c_1}(x-a) + \underbrace{\frac{1}{2!}f''(a)}_{c_2}(x-a)^2 + \underbrace{\frac{1}{3!}f'''(a)}_{c_3}(x-a)^3 \dots$$

This is the Taylor series for the function $f(x)$ at a. In the special case when $a = 0$, we also call this the Maclaurin series.

We have derived the conditions: $c_0 = f(0), c_1 = f'(0) = \frac{1}{1!}f'(0), c_2 = \frac{1}{2}f''(0) = \frac{1}{2!}f''(0)$ Try to show $c_3 = \frac{1}{6}f'''(0) = \frac{1}{3!}f'''(0)$ for yourself

A single point holds all the information

What do we need to define a function?



This also works in higher dimensions

$$f(\mathbf{x}) = f(\mathbf{x}_0) + \mathbf{A}^T(\mathbf{x} - \mathbf{x}_0) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)^T \mathbf{H}(\mathbf{x} - \mathbf{x}_0) + \dots$$

where $\mathbf{A} = \nabla_{\mathbf{x}} f(\mathbf{x})|_{\mathbf{x}_0}$ and $\mathbf{H}$ is the Hessian with $H_{ij} = \frac{\partial^2}{\partial_{x_i} \partial_{x_j}} f(\mathbf{x})\Big|_{\mathbf{x}_0}$

Reading

Recommended Readings

James, Witten, Hastie & Tibshirani, *An Introduction to Statistical Learning*

- Ch. 3.1, p. 61 - 71 - simple linear regression, its coefficients, and their accuracy
- Ch. 3.2, p. 71 - 75 - multiple linear regression until 3.2.2
- Ch. 3.2, p. 75 - 83 - remainder of Ch. 3.2
- Ch. 6.2, p. 237 - 251 - ridge regression and the lasso

Additional Resources

Richard McElrath, *Statistical Retinking*

- Ch.4 p.71 -122
- see also video lecture A03 from Statistical Retinking 2026

3Blue1Brown

- Taylor series - Chapter 11, Essence of calculus